Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/16/2024
PI Name:	Crucian, Brian Ph.D.		
Project Title:	Hazards of Lunar Surface Exploration: Determining the Immunogenicity/ Allergenicity of Lunar Dust		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-7061
Organization Name:	NASA Johnson Space Center		
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:		**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	10/01/2020	**End Date:**	09/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per K. Basso/JSC (Ed., 10/6/22) NOTE: End date changed to 09/30/2022 per K. Lehnhardt/JSC (Ed., 5/6/22)		
Key Personnel Changes/Previous PI:	August 2022 report: Audrie Colorado, Ph.D. was added as Co-Investigator. She replaced George Makedonas, Ph.D. on the investigation, who departed NASA service.		
COI Name (Institution):	Krieger, Stephanie M.S. (KBR/NASA Johnson Space Center) Marshall, Gailen Ph.D. (University of Mississippi Medical Center) Mercaldo, Nathaniel Ph.D. (KBR/NASA Johnson Space Center) Nelman-Gonzalez, Mayra B.A. (KBR/NASA Johnson Space Center) Colorado, Audrie Ann (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Lunar dust (LD) is unique. Due to lack of water or wind erosion, it consists of very sharp microscopic particles which stick to skin, clothing, and equipment, and are easily inhaled. During the Apollo moon missions, there were consistent reports of lunar dust exposure leading to upper respiratory symptoms in both astronauts and ground support personnel. In some cases, repeated exposure led to worsening symptoms. There could be several mechanisms by which LD causes symptoms, including simple particulate irritation of mucus membranes, acting as a toxin, or by activating the immune system to inflammation (allergy). International Space Station (ISS) crews already display a pattern of immune system problems that predispose them to both inflammation and allergy symptoms. It is essentially uninvestigated, if LD can serve as an allergen, or otherwise 'stimulate' the immune system to achieve symptoms. The goal of this study is to determine, in a simple cell culture experiments, if LD has the capacity to serve as an allergen, an adjuvant (something that increases allergy to other 'common' allergens), or a cellular toxin. We will obtain living immune cells from the blood of healthy test subjects. The immune cells will be purified, and then cultured in the presence of common allergens with or without lunar dust. We will then measure various factors to determine if LD caused any type of response from the immune cells such as activation or allergy. The measured factors include histamine, Immunoglobulin E (IgE, the main protein that 'triggers' allergy), or activation of basophils (the cell which causes allergy). There is currently only anecdotal data regarding human health, allergy, and LD exposure. The data from this study will inform how NASA responds if an astronaut experiences symptoms on the lunar surface. The data will also influence operations and/or vehicle engineering design for dust containment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will advance our understanding of allergenic responses in healthy and atopic individuals.
Task Progress:	This study is complete and a final report was sent to the NASA center admin in Sept 2023. An oral presentation was given at the NASA Human Research Program (HRP) Investigators' Workshop (IWS) 2024. The presentation may be downloaded from the Cumulative Bibliography of this record. This study aimed to investigate the potential of allergy specific reactions to lunar dust (LD), primarily looking at the key immune cell responsiveness to LD itself, a simulant, and other common recall antigens or polyclonal mitogens serving as positive controls. Human primary leukocytes isolated from blood were used for all assessments. As the primary mediators of allergy (eosinophils and basophils) are somewhat rare-percentages in the peripheral blood of healthy individuals, two cell lines representing those populations were also obtained and utilized. Cellular responsiveness to co-culture in the presence of LD was evaluated in several different ways. In this investigation, assessments in primary human subject blood immune cells indicated no evidence for sensitization, cellular responsiveness, nor 'allergy' to LD.
Bibliography Type:	Description: (Last Updated: 05/21/2026)
Articles in Peer-reviewed Journals	Colorado AA, Gutierrez CL, Nelman-Gonzalez M, Marshall GD, Mccoy JT, Crucian BE. "Hazards of lunar surface exploration: Determining the immunogenicity/allergenicity of lunar dust." Front Immunol. 2025 May 8;16:1539163. https://doi.org/10.3389/fimmu.2025.1539163 , May-2025
Articles in Peer-reviewed Journals	Crucian BE, Quiriarte H, Lam CW, Nelman M, Colorado AA, Diak DM, James JT. "Pulmonary and systemic immune alterations in rats exposed to airborne lunar dust." Front Immunol. 2025 Feb 5;16:1538421. https://doi.org/10.3389/fimmu.2025.1538421 ; PMID: 39981230; PMCID: PMC11840967 , Feb-2025
Conference Materials (Downloadable)	Colorado AA. "Hazards of lunar surface exploration: Determining the immunogenicity/allergenicity of lunar dust." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-15, 2024. , Feb-2024 IWS 2024.pptx